Day 5 —SPL Cheat Sheet — Splunk Free

Name:	Mohammed Farhan
Role:	Associate – Information Security (Intern)
Organization:	CyberShelter UAE
Date:	1 July 2026

Environment

Source	**Sourcetype**	**Method**
Linux auth.log	linux_auth	Continuous monitoring (live)
Windows Security + Sysmon	WinEventLog:Security / WinEventLog:Microsoft-Windows-Sysmon/Operational	Universal Forwarder (live, real-time)
Kali firewall log	ufw_firewall	Static one-time snapshot

Part 1 — Splunk Installation on Ubuntu Server

Downloaded Splunk :

- `Wget-Osplunk.deb "https://download.splunk.com/products/splunk/releases/10.4.1/linux/splunk-10.4.1-5a009d941268-linuxamd64.deb"`

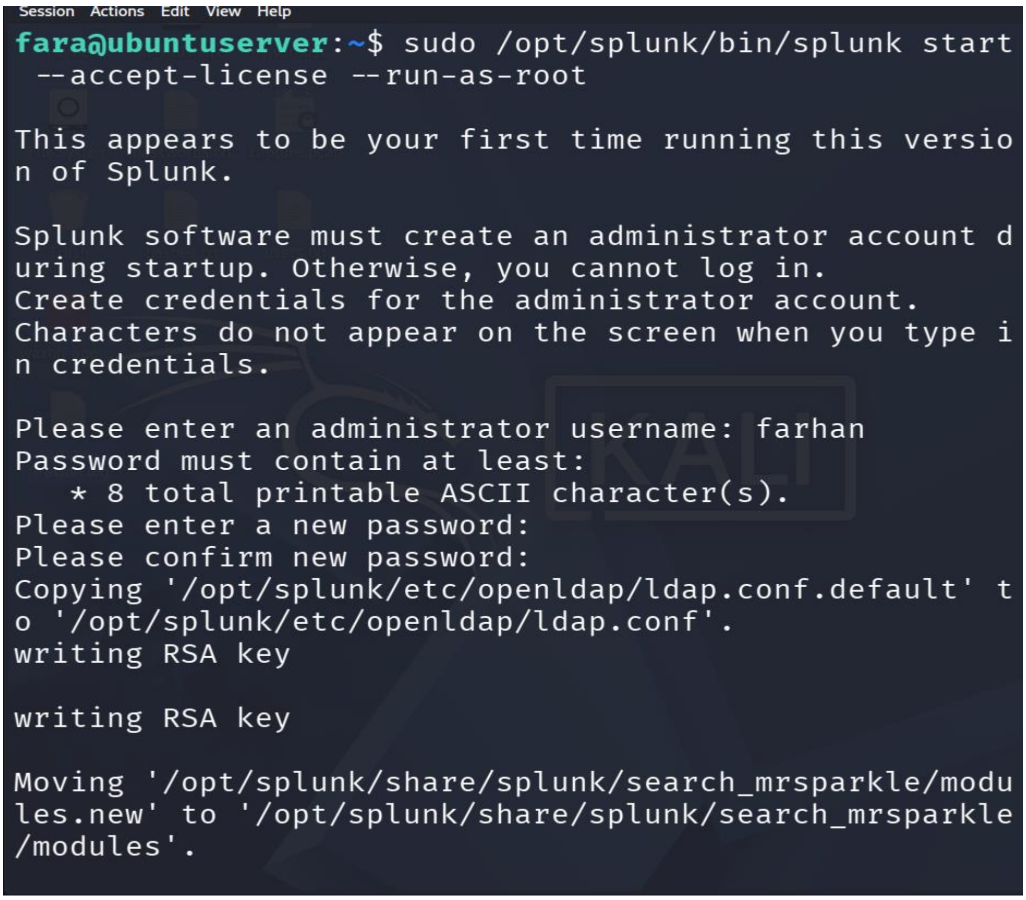
Installed:

- `sudo dpkg -i splunk.deb`

```
fara@ubuntuserver:~$ wget -O splunk.deb "https://download.splunk.com/products
/splunk/releases/10.4.1/linux/splunk-10.4.1-5a009d941268-linux-amd64.deb"
--2026-07-03 13:53:46--  https://download.splunk.com/products/splunk/releases
/10.4.1/linux/splunk-10.4.1-5a009d941268-linux-amd64.deb
Resolving download.splunk.com (download.splunk.com)... 108.139.86.27, 108.139
.86.34, 108.139.86.117, ...
Connecting to download.splunk.com (download.splunk.com)|108.139.86.27|:443...
 connected.
HTTP request sent, awaiting response... 200 OK
Length: 1322956776 (1.2G) [binary/octet-stream]
Saving to: ‘splunk.deb’

splunk.deb          100%[================>]   1.23G   720KB/s    in 45m 10s

2026-07-03 14:38:58 (477 KB/s) - ‘splunk.deb’ saved [1322956776/1322956776]

fara@ubuntuserver:~$ ls -lh splunk.deb
-rw-rw-r-- 1 fara fara 1.3G Jun 26 01:08 splunk.deb
fara@ubuntuserver:~$
```

Started Splunk (needed `--run-as-root` since running as root is blocked by default):

- `sudo /opt/splunk/bin/splunk start --accept-license --run-as-root`



```
Session  Actions  Edit  View  Help
fara@ubuntuserver:~$ sudo /opt/splunk/bin/splunk start
 --accept-license --run-as-root

This appears to be your first time running this versio
n of Splunk.

Splunk software must create an administrator account d
uring startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type i
n credentials.

Please enter an administrator username: farhan
Password must contain at least:
   * 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
Copying '/opt/splunk/etc/openldap/ldap.conf.default' t
o '/opt/splunk/etc/openldap/ldap.conf'.
writing RSA key

writing RSA key

Moving '/opt/splunk/share/splunk/search_mrsparkle/modu
les.new' to '/opt/splunk/share/splunk/search_mrsparkle
/modules'.
```

Verified running:

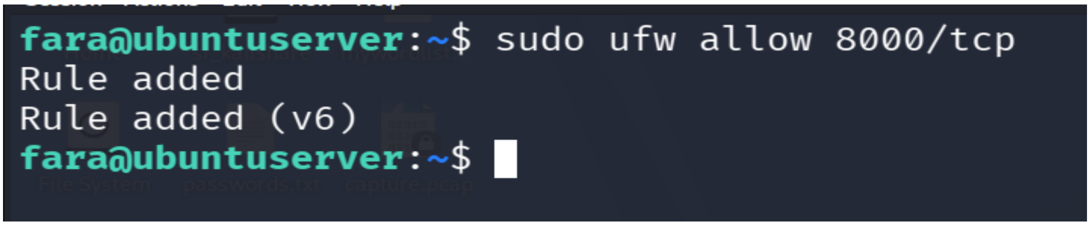
- `sudo /opt/splunk/bin/splunk status`

Confirmed splunkd running, web interface on port 8000.

- `sudo ss -tulnp | grep splunk`

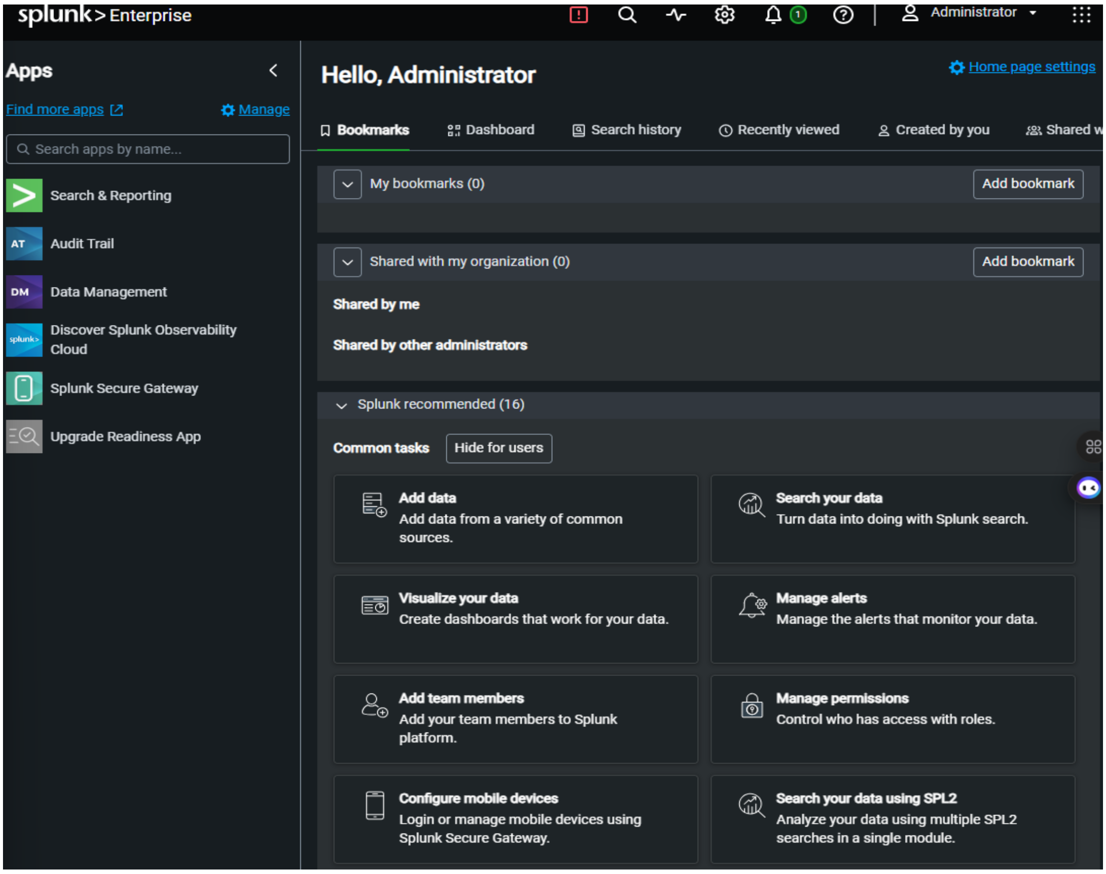
Opened firewall port:

- `sudo ufw allow 8000/tcp`



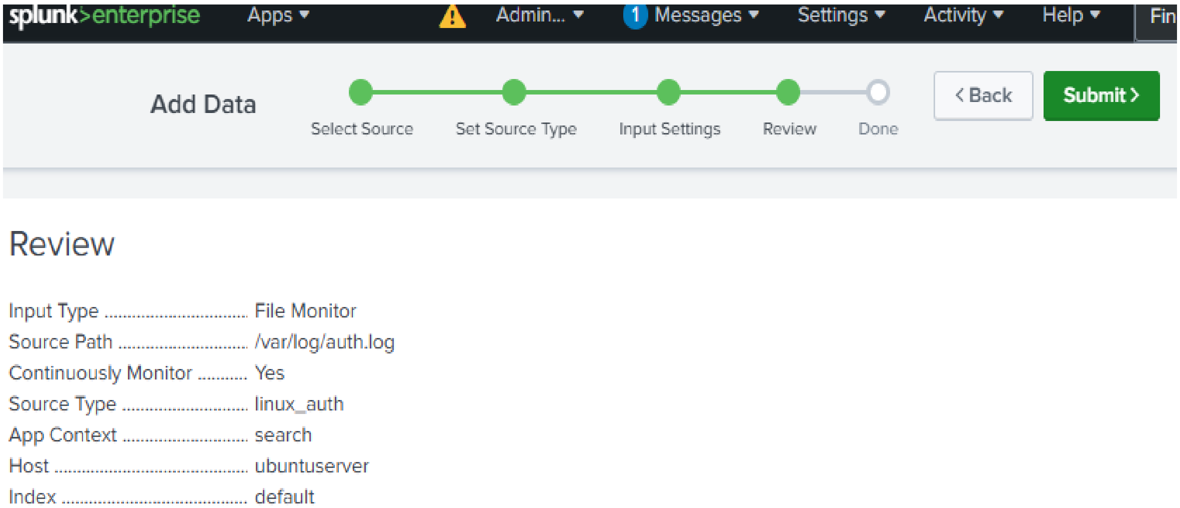
Accessed and logged in:

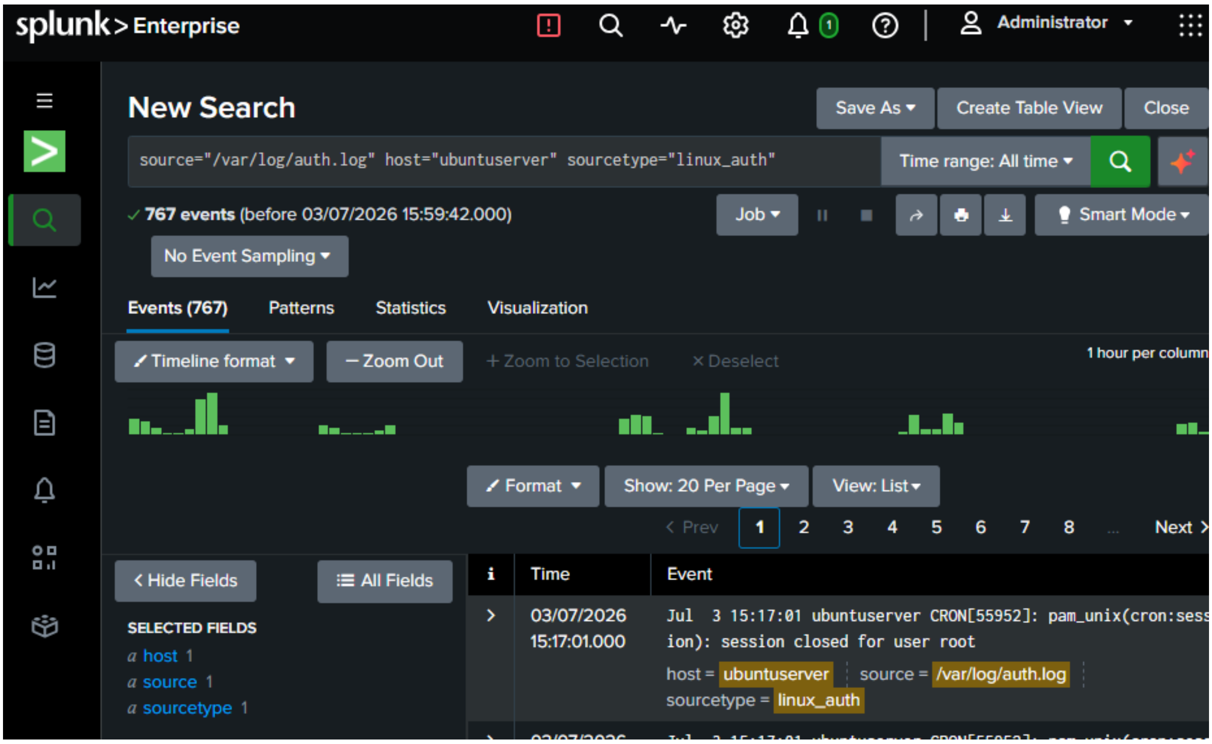
- http://192.168.56.104:8000



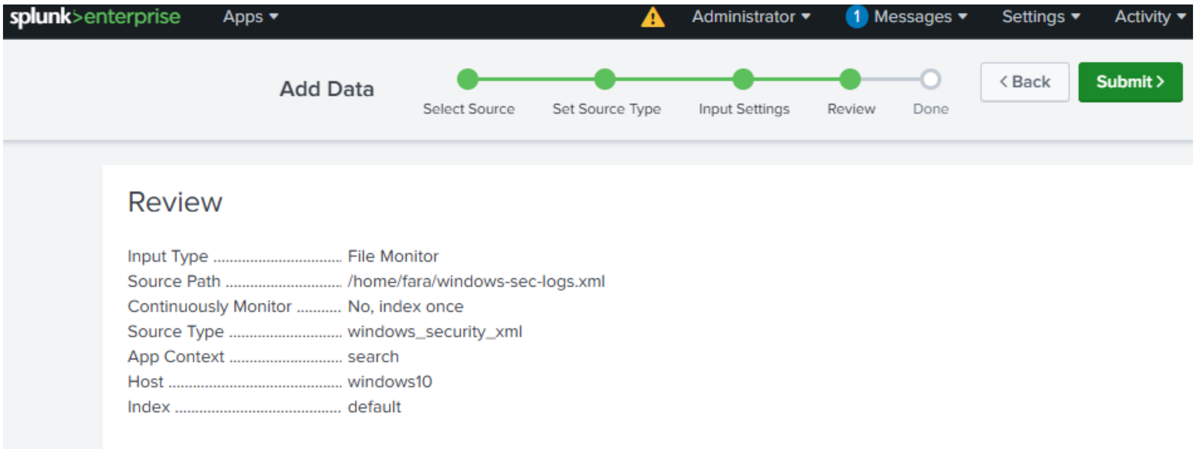
Part 2 — Log Source Ingestion (Initial Static Approach)

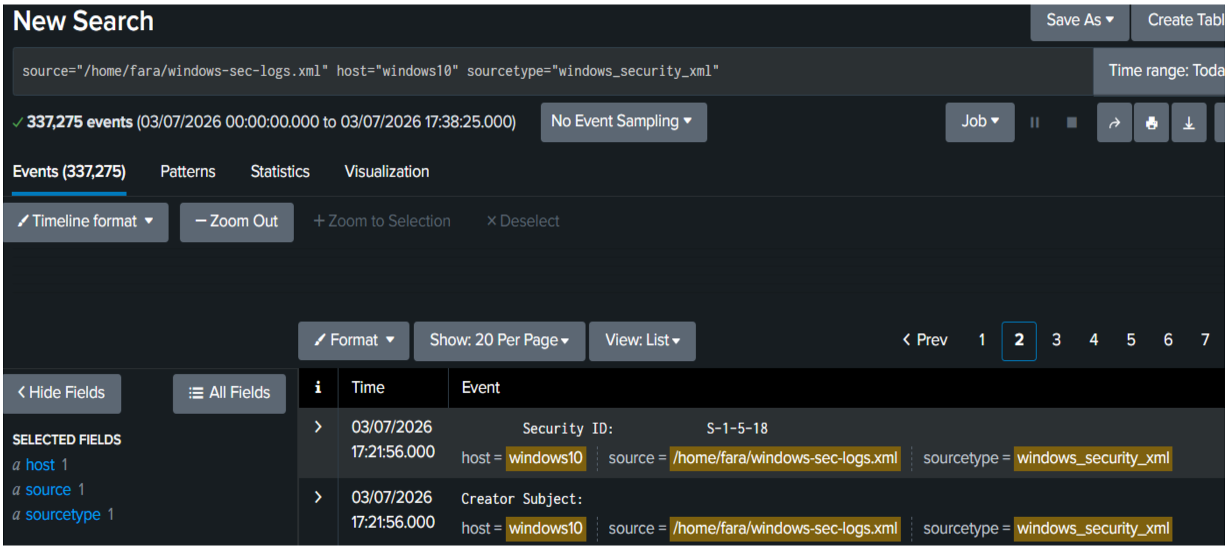
Source 1 — Linux auth.log: Monitored directly on Ubuntu Server via Splunk's "Add Data → Monitor → Files & Directories," continuously monitoring `/var/log/auth.log`, sourcetype `linux_auth`.



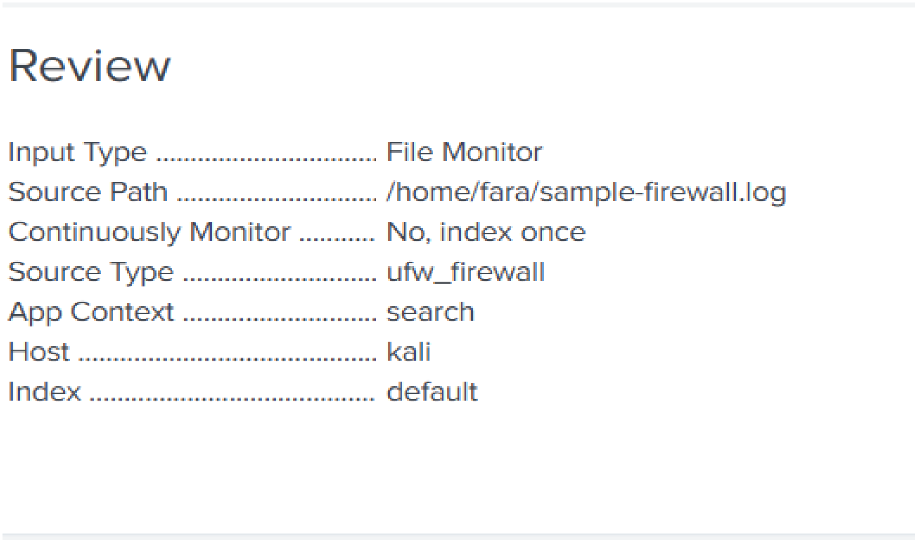


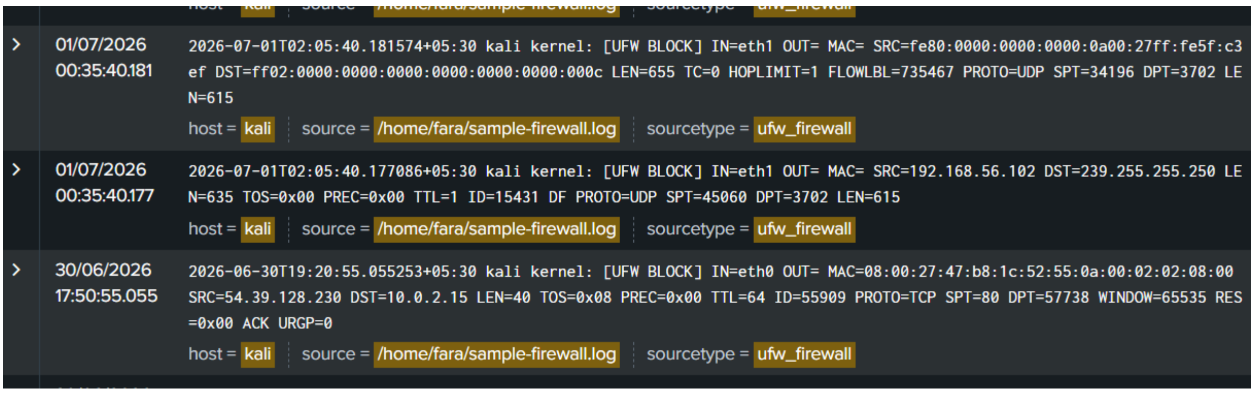
Source 2 — Windows Security XML: Generated test events on Windows 10 (failed login, PowerShell execution, new user creation, group membership change), exported Security log via Event Viewer as `windows-sec-logs.xml`, transferred via SCP to Ubuntu Server, ingested as one-time index with sourcetype `windows_security_xml` (log4net_xml parsing, host=windows10).





Source 3 — Firewall log: After multiple failed attempts to find a working public sample dataset, used Kali's own real UFW log (`/var/log/ufw.log`) — copied to Desktop, transferred via SCP to Ubuntu Server, ingested as one-time index, sourcetype `ufw_firewall`, host=kali.

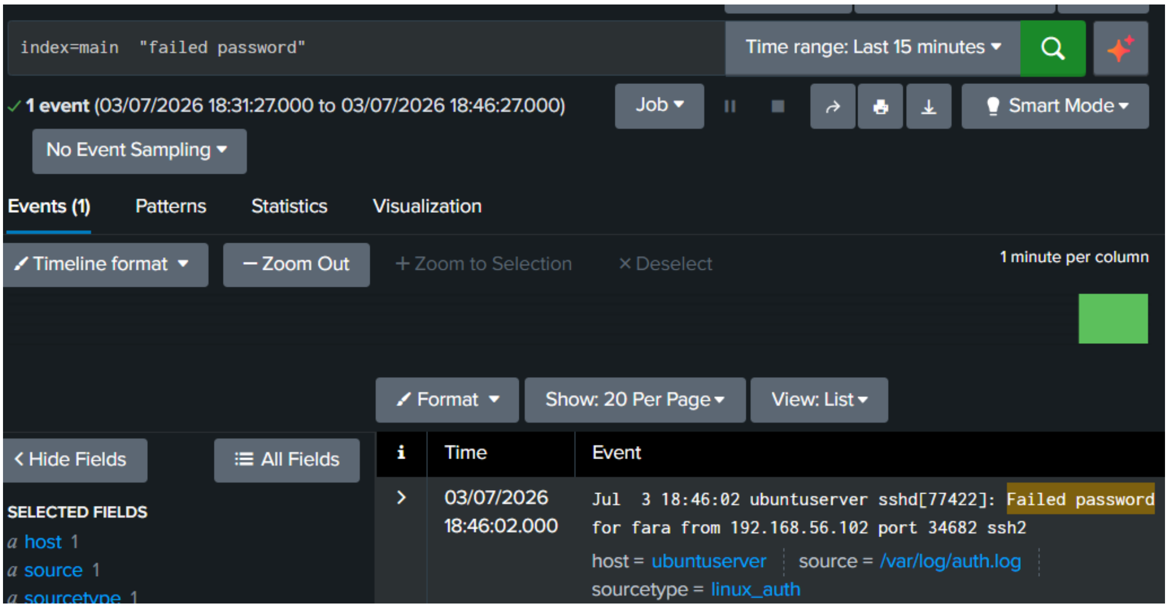




SPL Cheat Sheet

Search 1 — Login Activity (Failed & Successful)

spl
`index=main sourcetype=linux_auth "Failed password"`

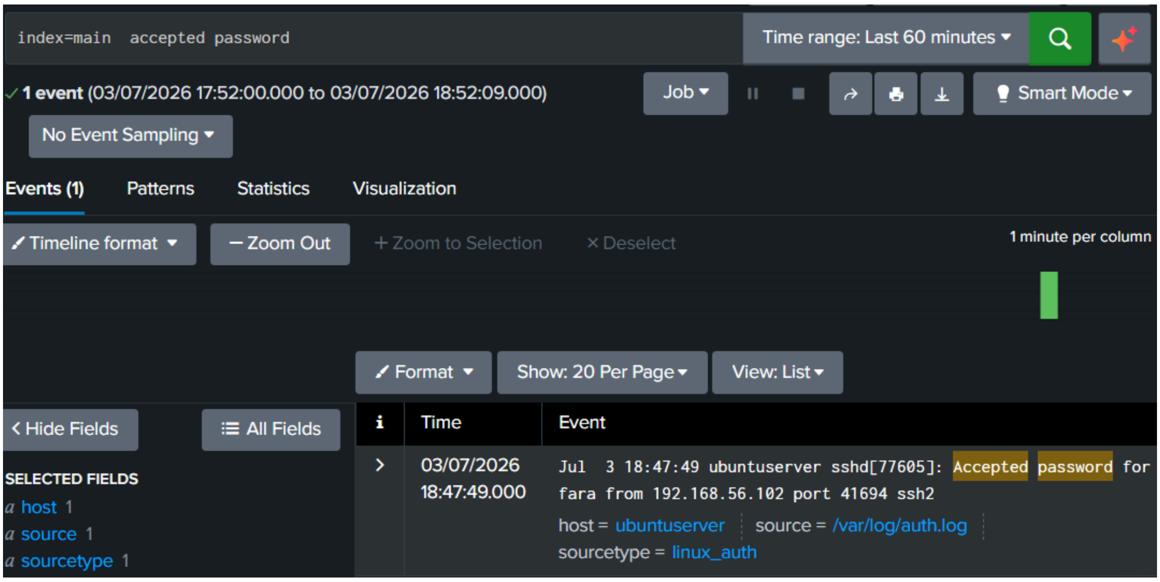


Detects: Brute force login attempts

MITRE ATT&CK: T1110 — Brute Force

spl

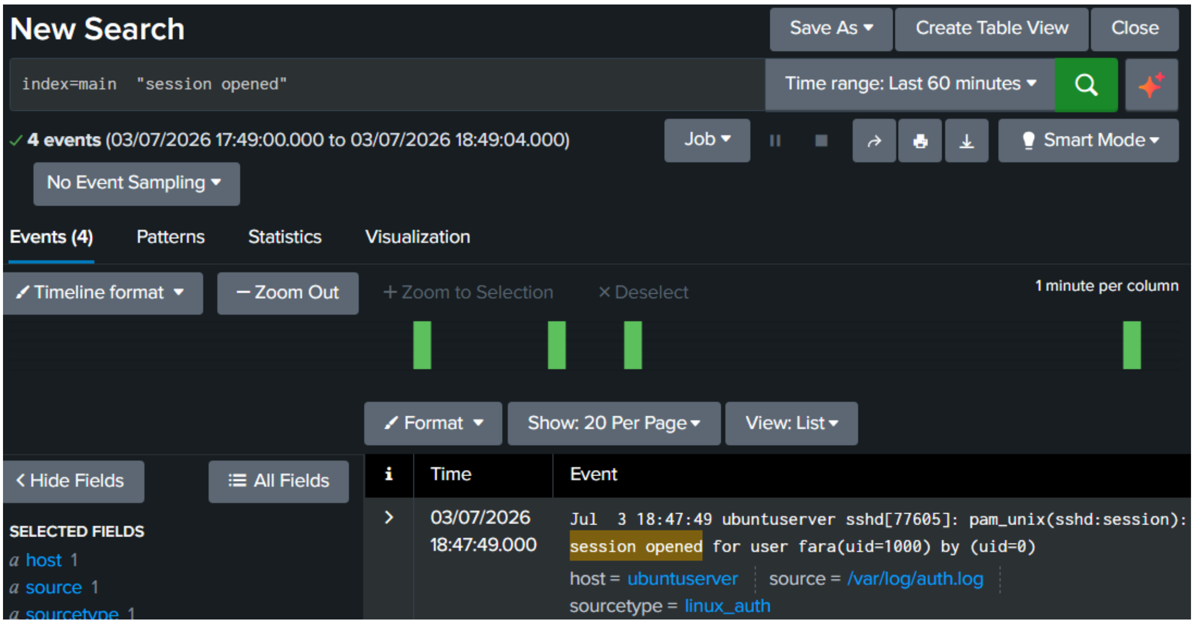
```
index=main sourcetype=linux_auth "accepted password"
```



Detects: Successful SSH logins

MITRE ATT&CK: T1078 — Valid Accounts

```
index=main sourcetype=linux_auth "session opened"
```



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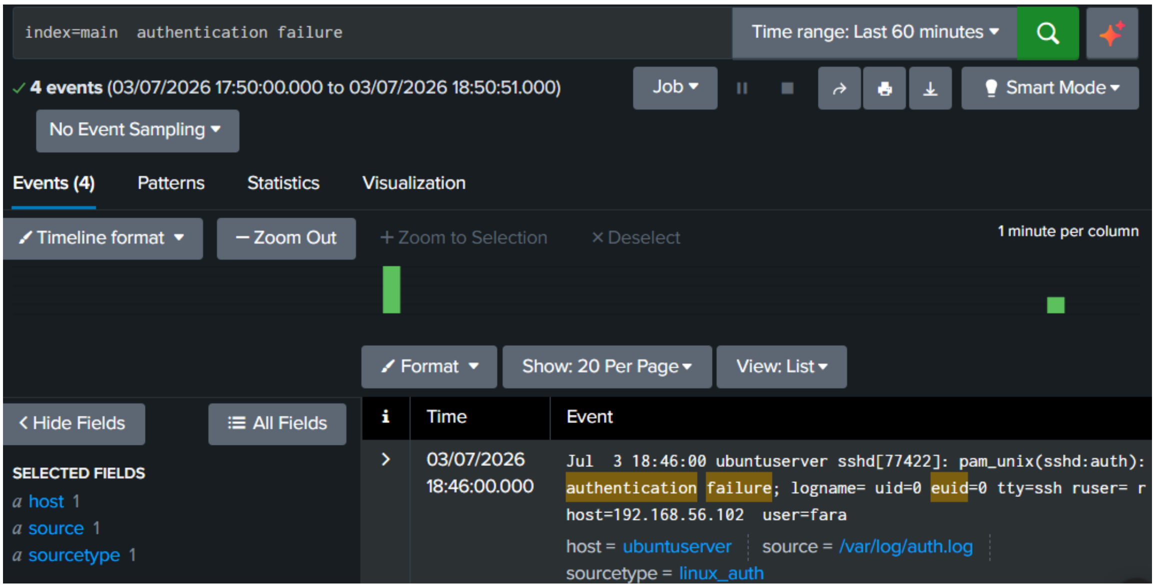
Detects: Baseline of all successful sessions — used to compare against failed attempts and spot anomalies

MITRE ATT&CK: T1078 – Valid Accounts

spl

```
index=main sourcetype=linux_auth "authentication failure"
```



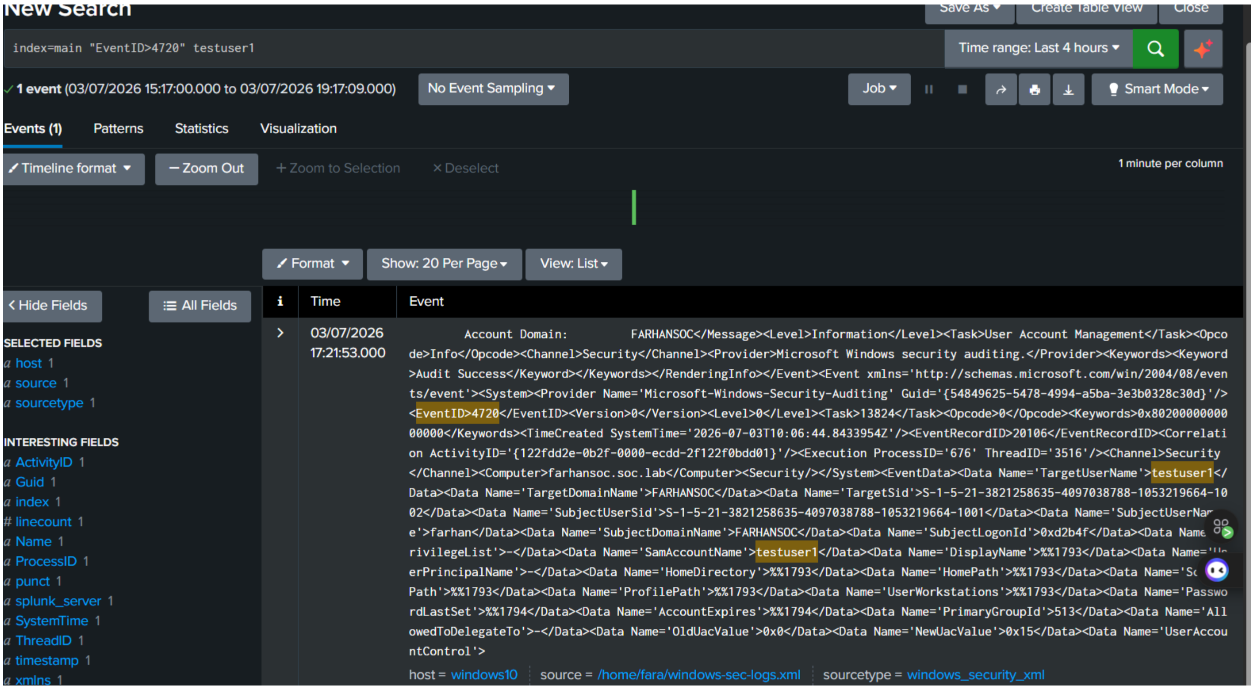
Detects: Broader PAM-level authentication failures

MITRE ATT&CK: T1110 — Brute Force

Search 2 — New User Account Creation

spl

`index=main "EventID>4720" testuser 1`

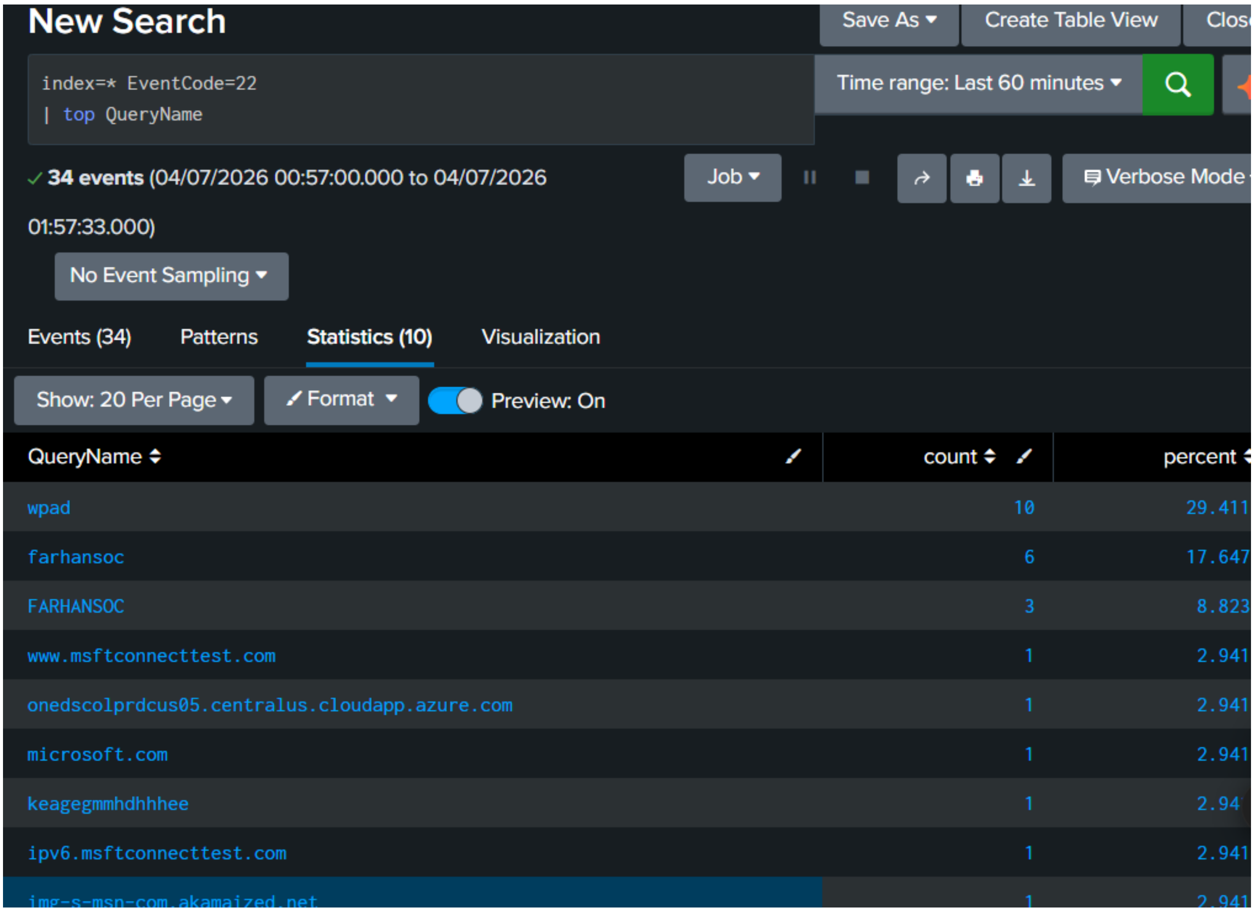


Detects: New user account created — attackers create backdoor accounts to maintain persistent access

MITRE ATT&CK: T1136 — Create Account

Search 4 — DNS Queries

spl
```
index=main EventCode=22 | top QueryName
```

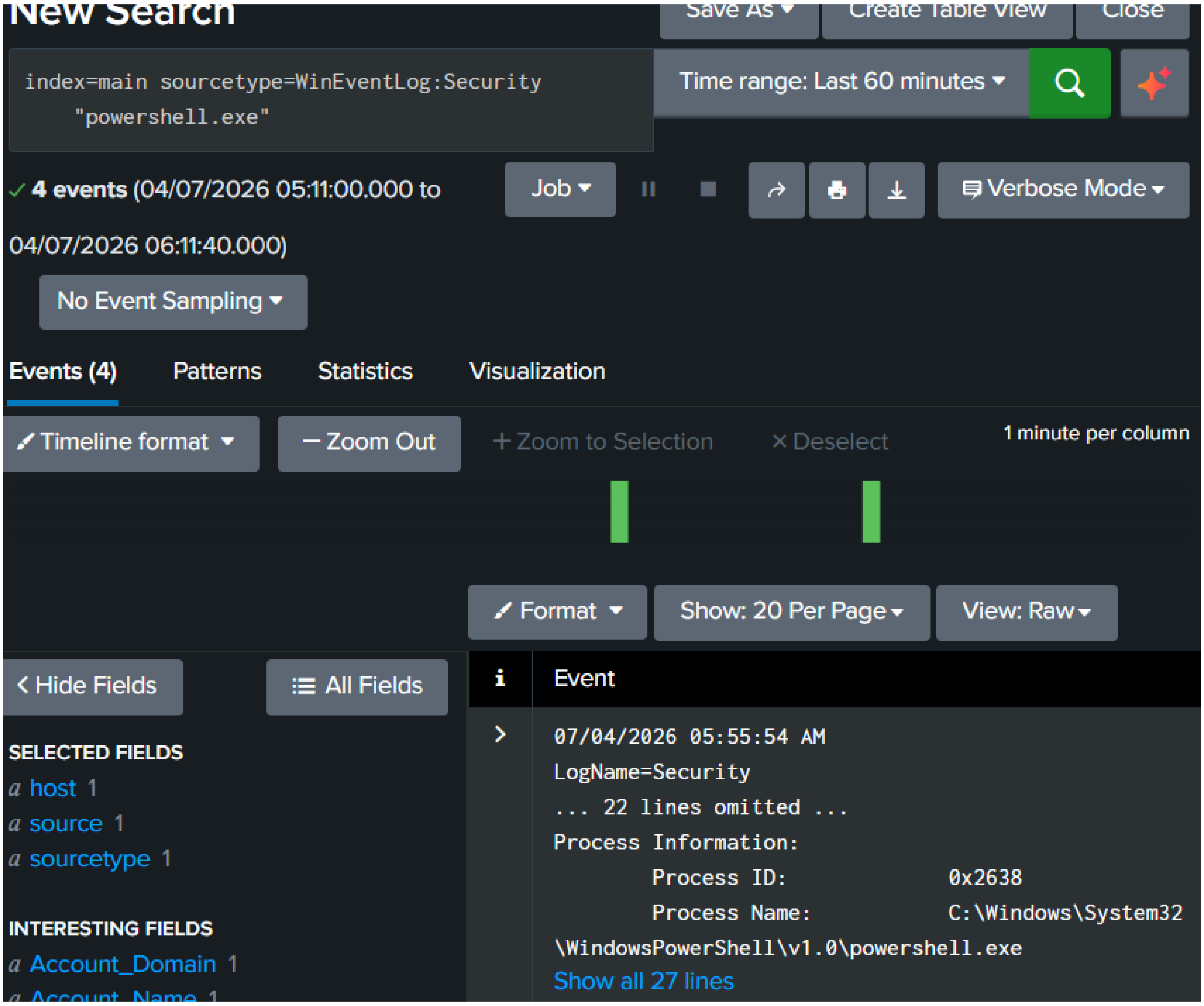


Detects: DNS queries made by processes, ranked by frequency — low-frequency/rare domains can indicate C2 (command and control) communication

MITRE ATT&CK: T1071.004 — Application Layer Protocol: DNS

Search 6 — PowerShell Executions

spl
```
index=main sourcetype=WinEventLog:Security "powershell.exe"
```



Detects: PowerShell execution — commonly abused for "living-off-the-land" attacks since it's a trusted, legitimate Windows tool

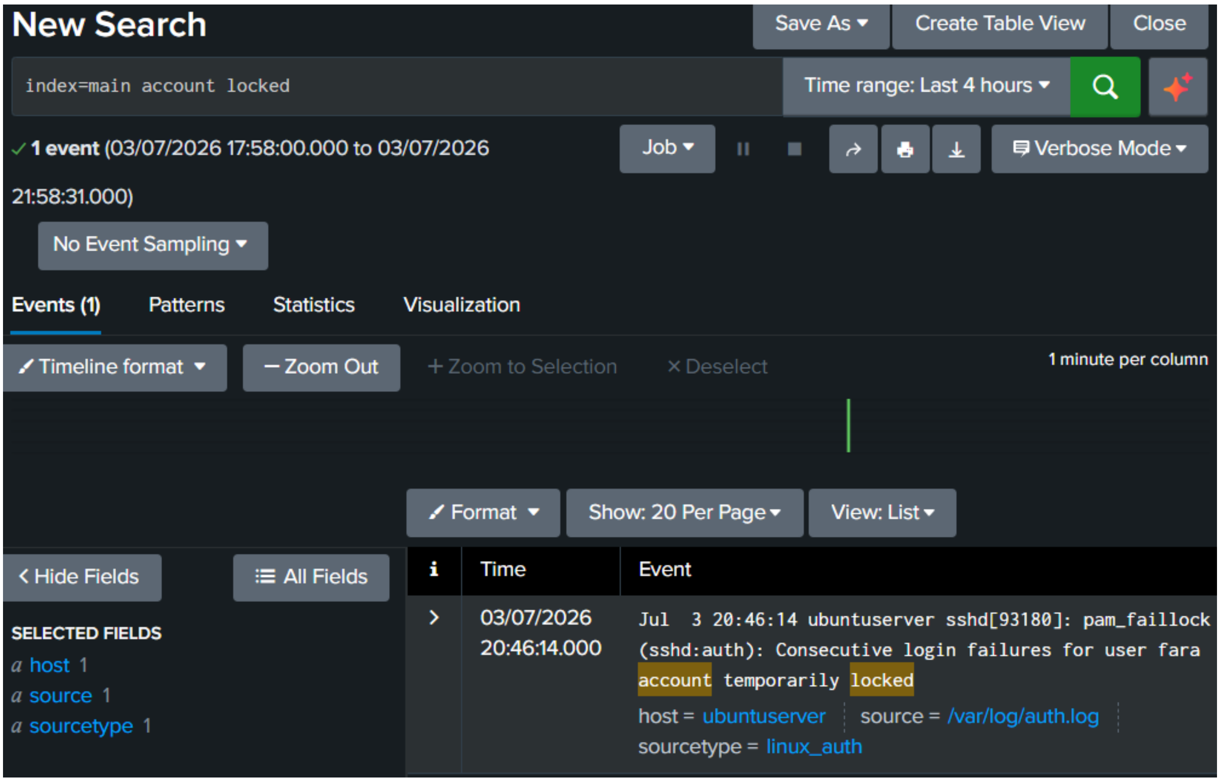
MITRE ATT&CK: T1059.001 — Command and Scripting Interpreter: PowerShell

Search 7 — Account Lockouts

Configuration: PAM faillock configured on Ubuntu Server (`/etc/pam.d/common-auth`, `/etc/pam.d/common-account`, `/etc/security/faillock.conf`) to lock accounts after repeated failed SSH attempts.

spl

```
index=main "account locked"
```



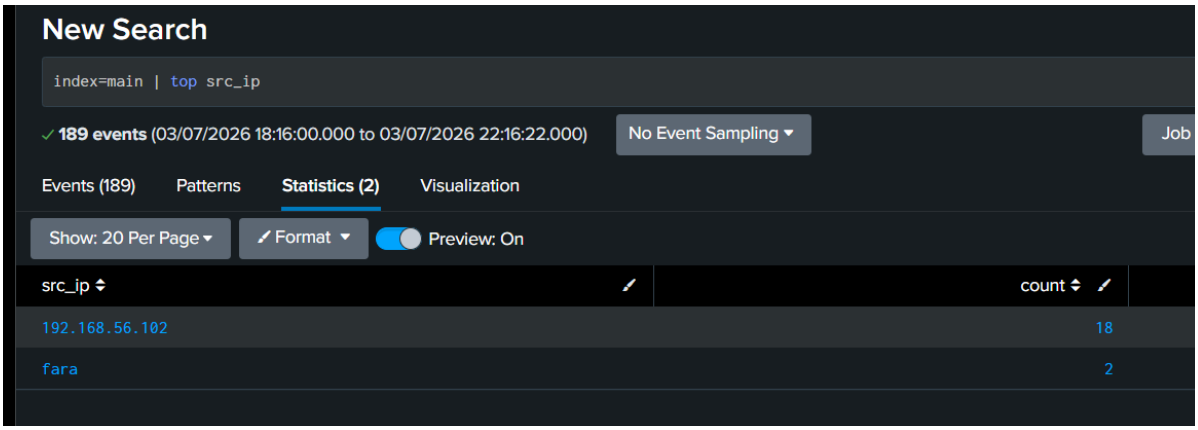
Detects: Account lockout triggered by repeated failed login attempts

MITRE ATT&CK: T1110 — Brute Force

Search 8 — Top Source IPs

spl

```
index=main | top src_ip
```

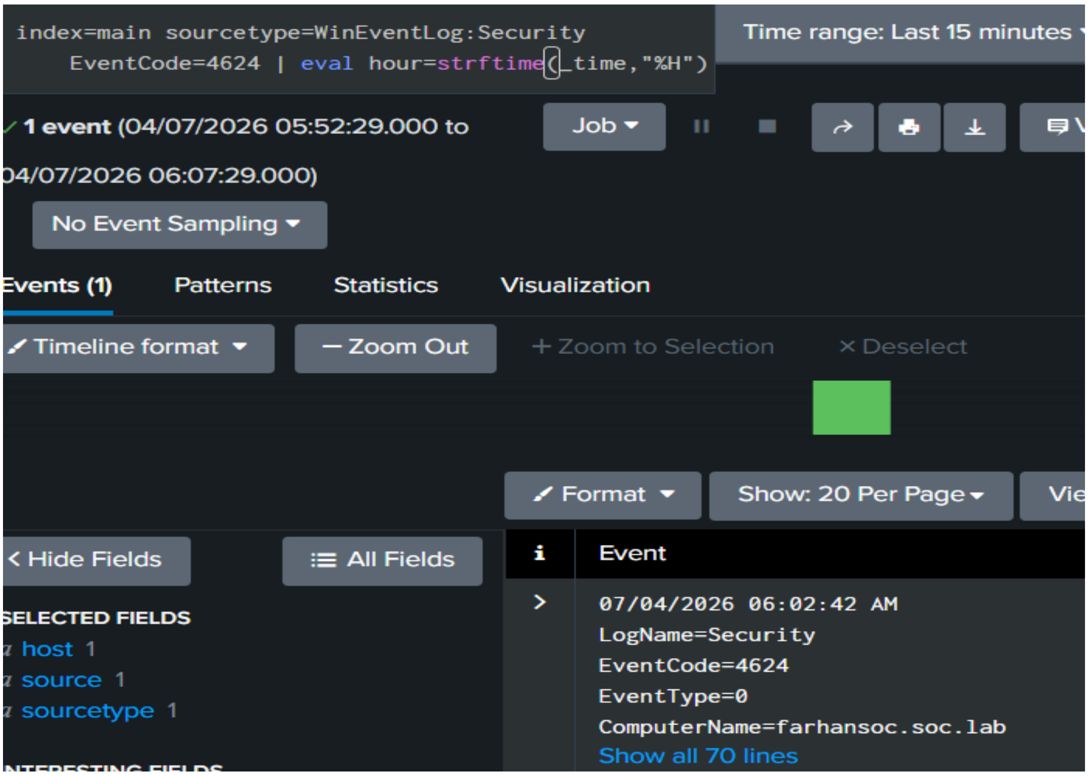


Detects: Most frequent source IPs connecting via SSH — identifies potential scanning or repeated brute force sources

MITRE ATT&CK: T1595 — Active Scanning

Search 9 — Logon Success Outside Business Hours

```
index=main sourcetype=WinEventLog:Security EventCode=4624 | eval hour=strftime(_time, "%H")
```



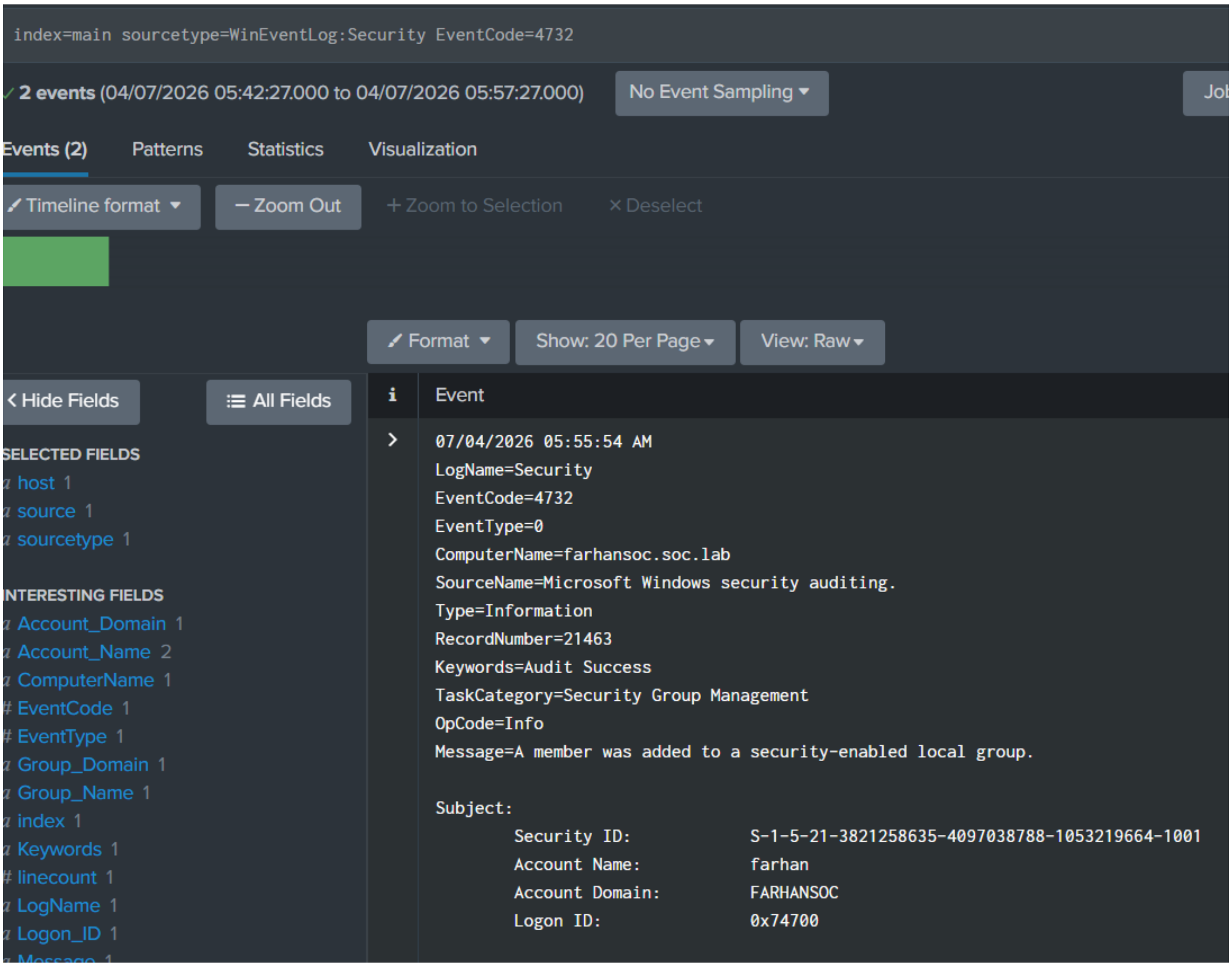
Detects: Successful logons occurring outside normal business hours (before 9 AM or after 6 PM) — indicator of compromised credentials being used at unusual times

MITRE ATT&CK: T1078 — Valid Accounts

Search 10 — Changes to Privileged Groups

spl

```
index=main sourcetype=WinEventLog:Security EventCode=4732
```



Detects: User added to a privileged/security-enabled group — classic privilege escalation technique

MITRE ATT&CK: T1098 — Account Manipulation

Conclusion

Eight of ten planned SPL searches were successfully built, tested, and confirmed with live data across Linux authentication logs, Windows Security events, and Sysmon telemetry. Real-world troubleshooting was required throughout the day. These fixes transformed static log snapshots into genuine live data pipelines, closely mirroring how log ingestion issues are diagnosed and resolved in real production SOC environments.